A 13-variable natural equation: source-only candidate

Nonnegative integer unknowns; 90 binary-operation abbreviations; exact degree 56 at the fixed ports.

The 13 unknowns below are distinct nonnegative integers:
`argJ, argFuel, argAlpha, argPellF, argIndex, argTime, argX`
`argScale, argOrdinate, argEta, argZeta, argAuxY, argGamma`
Each monospaced G-name below is an abbreviation, not an unknown. The fixed integers β, K, μ, ω are defined after the equation.

$\texttt{G001} = \texttt{argFuel} + 1$
$\texttt{G002} = \texttt{argAlpha} + 1$
$\texttt{G003} = \texttt{argIndex} + 1$
$\texttt{G004} = \texttt{argTime} + 1$
$\texttt{G005} = \texttt{argScale} + 1$
$\texttt{G006} = \texttt{argOrdinate} + 1$
$\texttt{G007} = \texttt{argEta} + 1$
$\texttt{G008} = \texttt{argZeta} + 1$
$\texttt{G009} = \texttt{argJ} \cdot \beta$
$\texttt{G010} = 1 + \texttt{G009}$
$\texttt{G011} = \texttt{G010} \cdot \texttt{G010}$
$\texttt{G012} = \texttt{G010} \cdot \texttt{G011}$
$\texttt{G013} = \texttt{G012} \cdot \texttt{G005}$
$\texttt{G014} = \texttt{G013} \cdot \texttt{G011}$
$\texttt{G015} = \texttt{G010} - \texttt{G001}$
$\texttt{G016} = \texttt{G015} - \texttt{G002}$
$\texttt{G017} = \texttt{argX} \cdot \texttt{G013}$
$\texttt{G018} = \texttt{G007} + \texttt{G008}$
$\texttt{G019} = \texttt{G013} \cdot \texttt{G018}$
$\texttt{G020} = \texttt{G007} + \texttt{G019}$
$\texttt{G021} = \texttt{G020} \cdot \texttt{G020}$
$\texttt{G022} = \texttt{G017} + \texttt{G013}$
$\texttt{G023} = 4 \cdot \texttt{G022}$
$\texttt{G024} = 3 + \texttt{G023}$
$\texttt{G025} = \texttt{G022} \cdot \texttt{G022}$
$\texttt{G026} = \texttt{G024} + \texttt{G025}$
$\texttt{G027} = \texttt{G022} \cdot \texttt{G020}$
$\texttt{G028} = \texttt{argX} + \texttt{G027}$
$\texttt{G029} = \texttt{G017} \cdot \texttt{G019}$
$\texttt{G030} = \texttt{argX} + K$
$\texttt{G031} = \texttt{G016} \cdot \texttt{G030}$
$\texttt{G032} = \texttt{G031} - \texttt{G001}$
$\texttt{G033} = \texttt{G029} + \texttt{G018}$
$\texttt{G034} = \texttt{G033} + \texttt{G006}$
$\texttt{G035} = \texttt{G018} \cdot \texttt{G032}$
$\texttt{G036} = \texttt{G034} - \texttt{G035}$
$\texttt{G037} = \texttt{G009} \cdot \texttt{G036}$
$\texttt{G038} = \texttt{G037} - \texttt{G032}$
$\texttt{G039} = \texttt{G038} \cdot \texttt{G038}$
$\texttt{G040} = \texttt{G009} \cdot \texttt{G009}$
$\texttt{G041} = \texttt{G029} \cdot \texttt{G033}$
$\texttt{G042} = 1 + \texttt{G041}$
$\texttt{G043} = \texttt{G040} \cdot \texttt{G042}$
$\texttt{G044} = \texttt{G039} - \texttt{G043}$
$\texttt{G045} = \texttt{G015} \cdot \texttt{G009}$
$\texttt{G046} = \texttt{G002} + \texttt{G045}$
$\texttt{G047} = \texttt{G011} - 1$
$\texttt{G048} = \texttt{G046} \cdot \texttt{G047}$
$\texttt{G049} = \omega \cdot \texttt{G010}$
$\texttt{G050} = \mu + \texttt{G049}$
$\texttt{G051} = \texttt{argJ} \cdot \texttt{G050}$
$\texttt{G052} = \texttt{G048} + \texttt{G051}$
$\texttt{G053} = \texttt{G018} - \texttt{G052}$
$\texttt{G054} = \texttt{G053} - 1$
$\texttt{G055} = \texttt{G014} \cdot \texttt{argGamma}$
$\texttt{G056} = \texttt{G017} \cdot \texttt{G010}$
$\texttt{G057} = \texttt{G055} + \texttt{G056}$
$\texttt{G058} = \texttt{G057} - \texttt{G054}$
$\texttt{G059} = \texttt{G014} \cdot \texttt{G028}$
$\texttt{G060} = \texttt{G024} \cdot \texttt{G058}$
$\texttt{G061} = \texttt{G059} + \texttt{G060}$
$\texttt{G062} = \texttt{G061} \cdot \texttt{G061}$
$\texttt{G063} = \texttt{G014} \cdot \texttt{G014}$
$\texttt{G064} = \texttt{G026} \cdot \texttt{G021}$
$\texttt{G065} = 1 + \texttt{G064}$
$\texttt{G066} = \texttt{G063} \cdot \texttt{G065}$
$\texttt{G067} = \texttt{G062} - \texttt{G066}$
$\texttt{G068} = \texttt{argPellF} \cdot \texttt{argPellF}$
$\texttt{G069} = \texttt{G068} - 1$
$\texttt{G070} = \texttt{G026} \cdot \texttt{G069}$
$\texttt{G071} = \texttt{G021} \cdot \texttt{G003}$
$\texttt{G072} = \texttt{G071} \cdot \texttt{G071}$
$\texttt{G073} = \texttt{G072} - \texttt{G070}$
$\texttt{G074} = \texttt{G004} \cdot \texttt{G020}$
$\texttt{G075} = \texttt{G074} - \texttt{G052}$
$\texttt{G076} = \texttt{G068} \cdot \texttt{G075}$
$\texttt{G077} = \texttt{G076} - \texttt{G020}$
$\texttt{G078} = \texttt{argAuxY} \cdot \texttt{argAuxY}$
$\texttt{G079} = \texttt{G077} \cdot \texttt{G077}$
$\texttt{G080} = \texttt{G078} - \texttt{G079}$
$\texttt{G081} = \texttt{G070} \cdot \texttt{G080}$
$\texttt{G082} = \texttt{G078} - \texttt{G081}$
$\texttt{G083} = \texttt{G044} \cdot \texttt{G044}$
$\texttt{G084} = \texttt{G067} \cdot \texttt{G067}$
$\texttt{G085} = \texttt{G082} - 1$
$\texttt{G086} = \texttt{G085} \cdot \texttt{G085}$
$\texttt{G087} = \texttt{G073} \cdot \texttt{G073}$
$\texttt{G088} = \texttt{G083} + \texttt{G084}$
$\texttt{G089} = \texttt{G086} + \texttt{G087}$
$\texttt{G090} = \texttt{G088} + \texttt{G089}$

The single equation is $\boxed{\texttt{G090} = 0}$.
The zero-map to the positive parent is an external standard-integer argument; its nonstandard model transfer and iPRA endpoint remain open.

The four fixed integers. All numerals, including the complete seed below, use ordinary base-10 notation. Put $\rho = 2^{67}$, $\beta = \rho^{8807} - 1$, and let $d_j = \lfloor P/10^{16j} \rfloor \bmod 10^{16}$ for $0 \le j < 943$. Let the ordered list $(\lambda_0, \ldots, \lambda_{40})$ be

$$1023, 1024, 1025, 1026, 1027, 1028, 1029, 1030, 1013, 1014, 1015, 1016, 1017, 1018,$$
$$1019, 1020, 1021, 1003, 1004, 1005, 1006, 1007, 1008, 1009, 1010, 983, 984, 985,$$
$$986, 987, 988, 989, 990, 991, 1031, 1022, 1011, 992, 1095, 2195, 3296$$

Define

$$\begin{aligned} K = {} & \rho^{30} + \rho^{62} + \rho^{1100} + \rho^{2201} + \rho^{8849} \\ & + \sum_{j=0}^{942} d_j (\rho^{4393-j} + \rho^{5495-j}) \\ & + \sum_{j=0}^{40} 2^{10+j} (\rho^{4393-\lambda_j} + \rho^{5495-\lambda_j}). \end{aligned}$$

For integer intervals $[a, b] = \{a, a+1, \ldots, b\}$, let

$$\begin{aligned} S = {} & ([0, 942] \cup [983, 992] \cup [1003, 1011] \cup [1013, 1031] \\ & \cup \{1063, 1064, 1095, 2195, 3296\}) \setminus \{1\}. \end{aligned}$$

Put $\mu = \beta - \sum_{k\in S} \rho^k - 2\rho^{1063}$ and $\omega = \beta + H$, where

$$H = 4 + \sum_{k=1013}^{1022} \rho^k + \sum_{k=1065}^{1073} \rho^k + \rho^{2195} + \rho^{3296} + (2^{51}-2)(\rho^{4393}+\rho^{5495}) + (2^{65}-1)\sum_{k=8792}^{8804} \rho^k + (2^{19}-1)\rho^{8805}.$$

Complete decimal integer P**.** Concatenate these lines in order, without spaces or line breaks:

17592072805377001759210268979300175921247098890017592127069185001759213204992100017592135195649001759214909030500017592101905490017557749245953001755774819840100175921081999370017557729850369001759214542643300175577455810570017557785426945001755778385510500175577282816010017592123409409001755775502131300175921522472970017557726187521001755777520844900175920977233930017592143337473001759212682342500175577749493770017592148058113001759217846681700175577587343370017592159070209001755779277926500175921055948810017557790684161001755779695615700175578043156490017592167462913001759214203596900175577264619530017592156979201001759217087283300175921596016650017592172708865001759217270886500175921732417700175578184775690017557792264193001759215279001700175921795295290017592177436010017557759238145001759211373465700175921142589450017592121074689001759215515340900175921603962890017592163017729001759218005708900175921805824010017592155941889001759216485478500175921779619850017592129466369001759218268262500175920982732810017592183733249001759207310950500175921643356170017592182947841001759218425958500175577990942730017557807482881001755781325004900175578174443530017557820590081001755782268723300175578237358090017557824260097001755782478438500175921231820810017557806959617001759217665945700175578171832330017592184785921000271441886003000027487788605450002851857814529000288621781504100029205772922890002954937292801000298929677004900030236567705610003058016247809000309237624832100031267357255690003161095726081000319545520332900032298152038410003264174681089000329853468160100033328941588400336072541593610003401613636609000343597363712100034703331143690003504693114881000353905259212900035734125926410003607772069889000364213207040100036764915476490003710851548161000374521102540900037795710259210003813930503169000384829050368100038826499809290003917099814410003951369458689000398572945920100040200889364490004054448936961000408880841420900041231684147210004157527891969000419188789248100042262473697290004260607370241000429496684748900043293268480010004363686325249000439804632576100044324058030090004460676580352100045011252807690004535485281281000456984475852900046042047590410004638564236289000467292423680100047072837140490004741643714561000477600319180900048103631923210004844722669569000487908267008100049134421473290004947802147841000498216162508900050165216256010005050881102849000508524110336100051196005806090005153960581121000518832005836900052226800588810005257039536129000529139953664100053257590138890005360119014401000539447849164900054288384921610005463197969409000549755796992100055319174471690005566277447681000560063692492900056349969254410005669356402689000570371640320100057380758804490005772435880961000580679535820900058411553587210005875514835969000590987483648100059442343137290005978594314241000601295379148900060473137920010006081673269249000611603326976100061503927470090006184752747521000621911222476900062534722252810006287831702529000632219170304100063565511802890006390911180801000642527065804900064596306585610006493990135809000652835013632100065627096135690006597069614081000663142909132900066657890918410006700148569089000673450856960100067688680468490006803228047361000683758752460900068719475251210006960307002369000694066700288100069750264801290007009386480641000704374595788900070781059584010007181184912385000721554491289700072499043901450007284264390657000731862386790500073529838684170007387343345665000742170334617700074560628234250007490422823937000752478230118500075591423016970007593501778945000762786177945700076622212567050007696581257217000773094073446500077653007349770007799660212225000783402021273700078683796899850007902739690497000793709916774500079714591682570008005818645505000804017864601700080745381232650008108898123777000814325760102500081776176015370008211977078785000824633707929700082806965565450008315056557057000841813551104100084524955115530008486854988801000852121498931300085555744665610008589934467073000862429394432100086586539448330008693013422081000872737342259300087617328998410008796092900353000883045237760100088648123781130008899171855361000893353185587300089678913331210009002251333633000903661081088100090707087113930009105330288641000913969028915300091740497664010009208409766913000924276924416100092771292446730009311488721921000934584872243300093802081996810009414568200193000948926776441000948328767795300095176471552010009552007155713000958636663296100096207266334730009655086110721000968944611123300097238055884810009758165588993000979252506624100098268850667530009929964020737000996432402124900999986834984970010033043499009001006740297625700101017629767690010136122454017001017048245452900102048419317770010239201932289001027356140953700103079214100490010342280887297001037664088780900104110003650570010445360365569004710479719842817001051407984332900105484393205770010582799321089001061715879833700106385766438500110981951416330011132555142145001116691461939300112012746199050011235634097153001126999409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The compact fixed-port recipe is distinct from the current-2011 equation. The external parent review is for standard integers and does not establish an iPRA endpoint for this expression.